# ADAPTIVE DIGITAL SAT PRACTICE TEST

# DSAT_Dec_2024_US_2-W

## 100% Original Compositions

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Section 1: Reading & Writing — Module 1 (27 Qs) | Module 2 Easier (27 Qs) | Module 2 Harder (27 Qs)

Section 2: Math — Module 1 (22 Qs) | Module 2 Easier (22 Qs) | Module 2 Harder (22 Qs)

Total: 148 Questions

# SECTION 1: READING & WRITING

## Module 1 (27 Questions — All Students)

### *Question 1*

Skill: Words in Context | Difficulty: Medium

The following text is adapted from Edwidge Danticat's 2004 novel *The Dew Breaker*. Ka is a young sculptor living in Brooklyn. In her studio, her favorite part of the morning routine, she <u>shapes</u> lumps of wet clay into angular faces with her thumbs. She presses dried flowers into wax tablets and <u>plants</u> tiny mosaic tiles along the borders of her picture frames.

As used in the text, what do the words "shapes" and "plants" both most nearly mean?

- A) Molds
- B) Grows
- C) Designs
- D) Positions

### *Question 2*

Skill: Words in Context | Difficulty: Medium

The intricate geometric patterns found on traditional Moroccan zellige tilework in Fez are iconic representations of North African artisanship. Artisans across the Mediterranean have adopted variations of this style — the mosaic facade of the Alhambra in Granada, Spain, for instance, is widely thought to _______ the decorative conventions of Moroccan zellige.

Which choice completes the text with the most logical and precise word or phrase?

- A) replicate
- B) imitate
- C) disregard
- D) surpass

### *Question 3*

Skill: Words in Context | Difficulty: Easy

In a warehouse, _______ the accumulation of excess inventory such as outdated electronics can be challenging because surplus goods tend to pile up rapidly and occupy valuable storage space, making it difficult to clear them entirely.

Which choice completes the text with the most logical and precise word or phrase?

- A) distributing
- B) cataloging
- C) curbing
- D) concealing

## *Question 4*

Skill: Words in Context | Difficulty: Medium

Specific design elements are nearly universal in the construction of pagodas, such as the tiered rooftop (or spire), which is regarded as a ________ of pagoda architecture. Even pagodas that blend features of several regional traditions, such as the Shwedagon Pagoda in Yangon, which draws from Burmese, Mon, and Indian influences, will still incorporate many of these conventional elements.

Which choice completes the text with the most logical and precise word or phrase?

A) trademark

B) contradiction

C) inspiration

D) replica

## *Question 5*

Skill: Text Structure and Purpose | Difficulty: Hard

Mariana Torres-Vega and her research team have examined how parallel adaptation — <u>a process in which organisms in geographically isolated populations independently develop similar traits — can arise from shared ancestral genetic variation or from entirely novel mutations</u>. Concurrently, Kenji Nakamura and colleagues have studied how parallel adaptation manifests across different environmental pressures, but the degree to which shared versus independent genetic pathways contribute to this phenomenon remains poorly characterized. This gap prompted biologists Rafael Mendes and Priya Chakraborty to assess both mechanisms of parallel adaptation in a comprehensive 2018 analysis.

Which choice best states the function of the underlined portion in the text as a whole?

A) It offers examples of how a phenomenon was investigated by researchers prior to Mendes and Chakraborty's analysis.

B) It provides a foundational description of a phenomenon that is central to the discussion that follows.

C) It clarifies a concept that the author suggests was misunderstood in the studies referenced in the text.

D) It introduces an experimental method that is explored in greater detail later in the text.

## *Question 6*

Skill: Text Structure and Purpose | Difficulty: Hard

In Quechua, an Indigenous language family spoken across the Andes region of South America, *wasi* means "house," whereas *wasiwasi* means "village" or "many houses." This linguistic device, in which part or all of a root word is repeated, sometimes with phonological alteration, within a derived word that is semantically related to the original, is known as reduplication. In this instance, the syllable *wasi* is doubled to form *wasiwasi*. There are numerous examples of this kind of reduplication in Quechua.

Which choice best describes the overall structure of the text?

A) It describes the relationship between Quechua and several other languages, raises a question about the nature of that relationship, and then answers that question.

B) It identifies the most commonly reduplicated words in Quechua, explains why it is difficult to translate those words into Spanish, and then provides examples of languages other than Spanish into which those words can be translated.

C) It presents some specific words in Quechua, describes the general linguistic phenomenon exemplified by those words, and then states that this phenomenon occurs frequently in Quechua.

D) It explains the phenomenon of reduplication, discusses why reduplication has been controversial among scholars, and then argues that an analysis of Quechua could help resolve that controversy.

### *Question 7*

Skill: Cross-Text Connections | Difficulty: Hard

**Text 1**

Henrik Larsen and his research group have identified geological evidence in sonar imagery of a 35-kilometer-wide subsurface depression beneath approximately one kilometer of sediment off the coast of northern Europe that is consistent with a 350-meter-wide celestial body striking the ocean floor. This formation, which the team designated Boreas, displays all the characteristic indicators of a high-velocity impact site: a raised perimeter, an elliptical outline, a stepped base, and a notable zone of central rebound.

**Text 2**

Both volcanic caldera collapse and deep-seated salt tectonics can produce crater-like depressions without requiring a high-velocity impact event. However, volcanic caldera formation is extremely improbable in the region near Boreas, and although salt tectonics could have plausibly occurred in this area and would produce a depression with a stepped base or an elliptical outline, it would not generate the zone of central rebound observed at Boreas.

Which choice best describes a difference between the approach of Text 1 and the approach of Text 2?

A) Text 1 objectively presents Larsen and colleagues' findings and conclusions, whereas Text 2 attempts to convey the researchers' enthusiasm upon discovering Boreas.

B) Text 1 focuses on features Boreas lacks, whereas Text 2 indicates features it shares with other geological depressions.

C) Text 1 discusses a single plausible cause of Boreas, whereas Text 2 evaluates two possible causes.

D) Text 1 emphasizes the evidence supporting a celestial impact as the cause of Boreas, whereas Text 2 argues against that explanation.

### *Question 8*

Skill: Inferences | Difficulty: Medium

The following text is from Nella Larsen's 1929 novel *Passing*. The narrator and her companion, daughters of a respected physician, are receiving instruction from Professor Aldridge.

Our instructor, Professor Aldridge, invited his younger colleague, David, to a lecture one afternoon. By then I was twenty and had absorbed everything Aldridge could impart, so I welcomed a fresh perspective. David was an emerging essayist, and he had learned from his colleague that the Aldridge sisters were none other than the daughters of Dr. Eloise Aldridge, and they possessed remarkable analytical minds. David was hoping not only to observe our seminar but to seek an introduction to the professor herself at Mother's residence.

Based on the text, why does David accompany his colleague to the lecture one day?

A) David has learned all his colleague can teach him and now desires to be taught by the sisters.

B) David has not received formal instruction in essay writing and wants to ask the sisters' famous mother to be his mentor.

C) David wants to present his essays to the sisters and inquire about their perspectives on his work.

D) David anticipates having the opportunity to be introduced to both the sisters and their mother.

### *Question 9*

Skill: Inferences | Difficulty: Hard

Determined to produce as many canvases as possible, Beatrice Palmer, a leading figure among the landscape painters known as the Carolina Naturalists, pioneered "rapid layering," which in part involved applying pigment across several surfaces simultaneously. That many of Palmer's followers, including Jerome Walsh, adopted the technique accounts in part for the impressionistic qualities that are now synonymous with the group's collective style. But not all Naturalists fully embraced this approach; for instance, though Clara Reeves was also prolific, her canvases were executed with greater precision and deliberation.

What does the text most strongly suggest about paintings by Walsh?

A) Because of the manner in which they were created, they likely have visual qualities that are regarded as more typical of Carolina Naturalist paintings than the qualities in works by Reeves are.

B) Although it is evident that Walsh adopted some of Palmer's preferred techniques, Walsh's works are less derivative of works by Palmer than is typically acknowledged.

C) The lack of precision with which they were executed suggests that they are inferior to works by either Palmer or Reeves.

D) Walsh's reliance on the technique of rapid layering likely accounts for his works being more aesthetically interesting than works by Reeves are.

### *Question 10*

Skill: Inferences | Difficulty: Hard

Copenhagen has high cycling rates, but other cities cannot increase their cycling rates simply by replicating a single feature of Copenhagen — e.g., its <u>extensive network of protected bike lanes</u> — that is associated with cycling culture. As urban planner Sofia Lindgren argues, many factors influence people's decision-making about whether to cycle: some studies have shown the importance of topography, others have shown the importance of climate and seasonal weather patterns, and so on, and it is clear that none of these factors in isolation fully explains cycling habits in a given city.

Based on the text, the author would most likely agree with which statement about Copenhagen's "extensive network of protected bike lanes"?

A) It may increase cycling in Copenhagen but is known to reduce cycling in other cities.

B) It is better understood as an effect of the high level of cycling in Copenhagen than as a cause of that cycling.

C) It affects cycling decisions in Copenhagen less than topography and seasonal weather patterns do.

D) It should be understood as just one of several factors that influence cycling activity in Copenhagen.

### *Question 11*

Skill: Command of Quantitative Evidence | Difficulty: Medium

Gabriela Fernandez and colleagues gathered data on 19 non-native shrub species cultivated in South America. They analyzed reports from Argentina, Chile, and Brazil about the number of these species grown in those countries as well as the numbers of beetle and mite species that damage those shrubs. The researchers concluded that Argentina had a greater number of damaging mite species than either of the other countries did.

[PLACEHOLDER: Table titled 'Numbers of the 19 Non-native Shrub Species Reported and the Beetle and Mite Threats to Them' with columns: Country | Shrubs | Mites | Beetles. Rows: Argentina: 14, 210, 95 | Brazil: 2, 12, 9 | Chile: 8, 31, 55]

Which choice best describes data from the table that support Fernandez and colleagues' conclusion?

A) Brazil reported 15 damaging mite species but only 9 damaging beetle species.

B) Argentina reported 210 damaging mite species, whereas Chile reported 55 damaging beetle species.

C) Argentina reported 210 damaging mite species, which is more than either Chile or Brazil reported.

D) Chile and Brazil reported 8 and 2 damaging mite species, respectively, which is far fewer than Argentina reported.

### *Question 12*

Skill: Textual Evidence | Difficulty: Hard

*Verses* is an 1897 collection of poetry by Paul Laurence Dunbar. In one of Dunbar's poems, the speaker criticizes those who vocally champion distant humanitarian causes while neglecting suffering in their own communities, saying _______

Which quotation from *Verses* most effectively illustrates the claim?

A) "The oak may fall before the storm / While reeds still bend and sway, / Yet strength alone will not transform / The darkness into day." (from "The Oak and the Reed")

B) "When ye weep for lands across the foam, / The stranger in his distant plight, / Remember those who starve at home / And perish in your city's night." (from "A Plea to the Comfortable")

C) "Sing praise to every hill and glen, / To rivers broad and free, / Where liberty shall rise again / For all humanity." (from "Ode to the Homeland")

D) "Let me craft the verses true, / Verses for the brave and bold; / Verses ringing sharp and new / Wherever courage may unfold." (from "Verses for the Brave")

### *Question 13*

Skill: Command of Quantitative Evidence | Difficulty: Medium

Many deciduous trees display leaves that are broader on one side of their central vein than the other, a condition known as foliar asymmetry. University of Michigan botanist Diane Kowalski and colleagues examined several species of maple and related trees, which produce leaves in opposite pairs, to determine if both leaves in a pair tend to exhibit asymmetry toward the same direction (same-direction asymmetry) or not. They found that opposite-direction asymmetry was far more common than same-direction asymmetry; in the sugar maple, for example, approximately ________

[PLACEHOLDER: Bar graph titled 'Asymmetry Direction of Leaf Pairs in Maples' with x-axis 'Asymmetry direction in pair' (opposite direction, same direction) and y-axis 'Number of pairs' (0 to 500). Three species shown: sugar maple (dark gray), red maple (light gray), silver maple (black). Opposite direction bars: sugar maple ~120, red maple ~130, silver maple ~480. Same direction bars: sugar maple ~50, red maple ~45, silver maple ~160.]

Which choice most effectively uses data from the graph to complete the example?

A) 120 leaf pairs show opposite-direction asymmetry, whereas approximately 50 pairs show same-direction asymmetry.

B) 85 leaf pairs show opposite-direction asymmetry, whereas approximately 35 pairs show same-direction asymmetry.

C) 500 leaf pairs show opposite-direction asymmetry, whereas no pairs show same-direction asymmetry.

D) 85 leaf pairs show opposite-direction asymmetry, whereas no pairs show same-direction asymmetry.

### *Question 14*

Skill: Inferences | Difficulty: Hard

Bluebell (*Hyacinthoides non-scripta*) plants are native to the British Isles, where temperate conditions have historically limited potential invasive competitors. As winters have become milder in recent decades, however, Spanish bluebell (*Hyacinthoides hispanica*) populations have expanded into Britain. It has been proposed that warming-induced extensions of the growing season in spring can benefit invasive species more than native ones; to evaluate this possibility, ecologists Thomas Whitfield and Anika Johansson tracked *H. non-scripta* and *H. hispanica*, along with other native and invasive species, over several years, concluding that invasives are advantaged by delays in frost onset during autumn in Britain.

Which finding, if true, would most directly support Whitfield and Johansson's conclusion?

A) Although *H. non-scripta* and *H. hispanica* both tended to extend their flowering period later into autumn in years with late frost onset, the extension was much greater for *H. hispanica* than for *H. non-scripta*.

B) Although significant year-to-year variations in frost onset were observed during the study, neither *H. hispanica* nor *H. non-scripta* showed any significant year-to-year variation in the timing of leaf senescence.

C) Although *H. non-scripta* and *H. hispanica* tended to cease flowering at about the same time in years with historically typical temperature patterns, *H. hispanica* ceased flowering later than *H. non-scripta* did in years with late frost onset.

D) Although *H. non-scripta* and *H. hispanica* both tended to produce more flowers overall in years with late frost onset than they did in years with historically typical temperature patterns, the years with late frost onset also had early growing season onset in spring.

### *Question 15*

Skill: Inferences | Difficulty: Hard

Rural Appalachian West Virginia's McDowell County is among the most sparsely populated counties in the eastern United States: the US Census Bureau classified it as 97.2% rural in 2010. Researchers studying populations of counties like McDowell often struggle to recruit and retain participants. Anthony Russo and colleagues tested whether a method called respondent-driven sampling could improve recruitment and retention. Working in two rural counties, the researchers identified a small number of people who had the characteristics desired for a proposed study and asked them to recruit additional participants from their personal networks. Russo and colleagues found that participants recruited via respondent-driven sampling showed a much higher retention rate than did people recruited by outsiders, suggesting that _______

Which choice most logically completes the text?

A) being recruited to participate in a study by someone with whom one is personally connected may create a sense of commitment to persist with participation in the study.

B) people with relatively small personal networks are inherently less likely to be recruited to participate in a study via respondent-driven sampling than are people with relatively large personal networks.

C) respondent-driven sampling is more likely to improve retention rates among rural participants than among nonrural participants.

D) personal networks can become large enough that two people can share a network but nevertheless regard each other as strangers.

### *Question 16*

Skill: Form, Structure, and Sense | Difficulty: Easy

*Portrait of a Musician* (1490) is one of several surviving works by Leonardo da Vinci's Milanese workshop. Currently, the painting _______ at the Pinacoteca Ambrosiana in Milan, where visitors can view the Italian Renaissance master's work on permanent display.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) will have hung

B) hangs

C) hung

D) was hanging

### *Question 17*

Skill: Form, Structure, and Sense | Difficulty: Medium

The human hand contains a complex arrangement of intrinsic muscles, each playing a vital role in the hand's dexterous movement. The flexor pollicis brevis, which is attached to the trapezium bone, _______ responsibility for bending the thumb at its base joint.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) bearing

B) to bear

C) bears

D) having borne

### *Question 18*

Skill: Form, Structure, and Sense | Difficulty: Easy

Georgia resident Tunis Campbell, one of the roughly two thousand African Americans elected to public office during the era that followed the Civil War, _______ his tenure as a state senator in 1868.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to begin
B) having begun
C) beginning
D) began

### *Question 19*

Skill: Boundaries | Difficulty: Medium

The ancient redwood (*Sequoia sempervirens*) known as Hyperion, located in northern California, was recognized as the tallest known tree on Earth, at 380 feet. With over two millennia of growth data in its _______ single tree like this, claims dendroclimatologist Patricia Engel, can reveal the ecological history of an entire region.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) rings, A
B) rings; a
C) rings and a
D) rings, a

### *Question 20*

Skill: Boundaries | Difficulty: Medium

After uncovering records about Robert Smalls, who represented South Carolina in the US House of Representatives, the historian discovered biographical accounts of two other Black Americans who served in _______ John Roy Lynch of Mississippi and Josiah Walls of Florida.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Congress;
B) Congress:
C) Congress.
D) Congress

## *Question 21*

Skill: Transitions | Difficulty: Medium

Marcus Whitman adopted the pen name "Aurelius" — the name of a celebrated Roman emperor and philosopher — in political pamphlets he composed in 1803, a choice that accomplished far more than simply disguising his identity. ________ it was a deliberate rhetorical strategy through which Whitman associated his political arguments with the esteemed civic ideals of the classical world, thereby strengthening the authority of his writing.

Which choice completes the text with the most logical transition?

A) Indeed,

B) Conversely,

C) In addition,

D) However,

## *Question 22*

Skill: Transitions | Difficulty: Medium

In Rachel Carson's *The Sea Around Us* — where, early on, the author marvels at the delicate bioluminescence of a single jellyfish but later recoils when contemplating vast colonies of jellyfish consuming entire fisheries — Carson grapples with the complicated dualities of the marine world. ________ the ocean's mesmerizing beauty and its ruthless indifference prove inseparable for Carson, like "two currents of the same tide."

Which choice completes the text with the most logical transition?

A) Ultimately,

B) To that end,

C) Hence,

D) Moreover,

## *Question 23*

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- The Andean condor is a species of bird that can be found in the Andes Mountains of South America.
- It has an average wingspan of 3.1 meters.
- The wandering albatross is a species of bird that can be found over the Southern Ocean.
- It has an average wingspan of 3.4 meters.

The student wants to emphasize a similarity between the Andean condor and the wandering albatross. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) On average, the Andean condor has a wingspan of 3.1 meters, while the wandering albatross has a wingspan of 3.4 meters.

B) The wandering albatross, which has an average wingspan of 3.4 meters, can be found over the Southern Ocean.

C) The wandering albatross and the Andean condor can both be found in the Southern Hemisphere.

D) The Andes Mountains of South America are home to several bird species, one of which is the Andean condor.

### *Question 24*

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

• The Beaufort scale is a system that classifies wind speed based on observed conditions at sea and on land.
• Winds at Beaufort number 3 are classified as a "gentle breeze" and blow at 12–19 km/h.
• Winds at Beaufort number 7 are classified as a "near gale" and blow at 50–61 km/h.
• Winds at Beaufort number 10 are classified as a "storm" and blow at 89–102 km/h.

The student wants to emphasize the wind speed of a near gale. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) On the Beaufort scale, a storm (number 10) has wind speeds of 89–102 km/h, which is faster than a near gale.

B) A near gale has a Beaufort number of 7, which means that it is stronger than a gentle breeze (number 3) but weaker than a storm (number 10).

C) The Beaufort scale classifies wind based on observed conditions; for example, a gentle breeze blows at 12–19 km/h.

D) Classified as Beaufort number 7, a near gale produces wind speeds of 50–61 km/h.

### *Question 25*

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

• Langston Hughes was a celebrated poet.
• His first published collection of poetry was a book.
• It was called *The Weary Blues*.
• It first appeared through Alfred A. Knopf in 1926.

The student wants to identify the title of Langston Hughes's first published poetry collection. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Langston Hughes's first published poetry collection appeared in 1926.

B) Langston Hughes's first published poetry collection was called *The Weary Blues*.

C) In 1926, a poetry collection by Langston Hughes appeared through Alfred A. Knopf.

D) Celebrated poet Langston Hughes's first published work of literature was a collection of poetry.

### *Question 26*

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

• Spatial orientation is a fundamental aspect of navigation.

• In the Montessori method of education, there is a sensory exercise for children called the stereognostic bag activity.

• In the activity, children identify objects inside a cloth bag using only touch.

• In traditional Waldorf education, there is a comparable exercise called form drawing.

• In form drawing, children trace geometric shapes in the air with their eyes closed.

The student wants to emphasize a similarity between the two exercises. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The stereognostic bag activity in the Montessori method and form drawing in Waldorf education are both exercises that develop children's spatial skills.

B) Both the stereognostic bag activity and form drawing are exercises found in major educational philosophies.

C) In the stereognostic bag activity, children identify objects by touch alone, in contrast to form drawing, where children trace shapes in the air with eyes closed.

D) In the Montessori method, there are a number of sensory exercises, including the stereognostic bag activity.

### *Question 27*

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

• *Equilibrium* is a 2020 sculpture by Brazilian artist Carlos Meireles.

• It is a suspended mobile constructed from recycled aluminum sheets.

• Mobiles became popular with artists in the 1930s and began to decline in mainstream popularity in the 1970s.

• Mobiles experienced a resurgence among artists in the 2000s and remain popular today.

• Sculptor Diana Varga notes, "[Mobiles] must be assembled by hand because there is no automated way to balance them."

The student wants to connect the sculpture to the history of mobiles in art. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Mobiles like *Equilibrium* are popular with artists and collectors, though they must, as Varga notes, be assembled by hand since they cannot be easily mass-produced.

B) With *Equilibrium*, Meireles joins the generations of artists who have helped sustain the mobile as an artistic medium since its resurgence in the 2000s.

C) Continuing from the 1930s to today, mobiles have been a popular artistic form, both as gallery installations and as kinetic sculptures.

D) The recycled aluminum mobile *Equilibrium* connects to the rich history of a medium whose artistic popularity has long been in decline.

# SECTION 1: READING & WRITING

## Module 2 — Easier Path (27 Questions)

(~70% Easy/Medium, ~30% Hard)

### *Question 1*

Skill: Words in Context | Difficulty: Easy

The following text is adapted from Mark Twain's 1876 novel *The Adventures of Tom Sawyer*. Tom is a schoolboy in a small Missouri town.

In the classroom, his least favorite part of the week, he <u>traces</u> letters of the alphabet on his slate with a worn piece of chalk. He folds scraps of paper into boats and <u>drops</u> them into the ink well on his desk.

As used in the text, what does the word "traces" most nearly mean?

- A) Follows
- B) Discovers
- C) Draws
- D) Investigates

### *Question 2*

Skill: Words in Context | Difficulty: Easy

Insights from historical declines in marine biodiversity can help inform conservation strategies aimed at protecting endangered aquatic species. The environmental pressures that led to the disappearance of the Caribbean monk seal around 1950 may also be _______ threatened marine mammals today.

Which choice completes the text with the most logical and precise word or phrase?

- A) hidden from
- B) relevant to
- C) combined with
- D) reliant on

### *Question 3*

Skill: Words in Context | Difficulty: Easy

In a 2019 article celebrating documentaries depicting Indigenous communities, reviewers for the *Los Angeles Times* _______ Nanook Revisited's 1975 film *Voices of the North* and Alanis Obomsawin's 1993 film *Kanehsatake*, praising the former as "deeply moving" and the latter as "unflinching and essential."

Which choice completes the text with the most logical and precise word or phrase?

- A) promoted
- B) parodied
- C) dismissed
- D) commended

### *Question 4*

Skill: Text Structure and Purpose | Difficulty: Medium

In 1885, in an effort to determine how long certain bacterial cultures could survive under laboratory conditions, microbiologist Heinrich Bauer sealed 15 glass vials containing nutrient broth and samples from various bacterial strains, including *Bacillus*, *Clostridium*, and *Staphylococcus*. Since then, some of the vials have been opened in periodic checks of viability that researchers are conducting. After 50 years had elapsed, most of the cultures in those vials had ceased to be viable, but *Bacillus* endospores have remained viable, including those from a vial opened in 2023 by microbiologists Sarah Lin, Jorge Medina, and colleagues.

Which choice best states the main purpose of the text?

A) To discuss an ongoing experiment

B) To explain how to eliminate certain bacteria

C) To compare two experiments addressing the same questions

D) To describe difficulties associated with preserving bacterial cultures

### *Question 5*

Skill: Text Structure and Purpose | Difficulty: Medium

Peruvian textile artist Luciana Calderon weaves narratives of endurance, using her distinctive method of intertwining metallic thread with photographs of people. In some works, Calderon focuses on honoring cultural figures who are people of Indigenous heritage, as she does in her depiction of activist Gregoria Apaza. However, in other works, Calderon celebrates everyday individuals, as she does in her luminous portrait of an elderly woman at a marketplace. Calderon sees both of these approaches as ways of illustrating the resilience and connectedness of all people.

Which choice best describes the overall structure of the text?

A) It introduces Calderon's cultural background, describes how this background influences her art, and then outlines some of her ideas for future projects.

B) It compares Calderon to other contemporary artists, indicates how two of her works are similar, and then emphasizes Calderon's enthusiasm for artistic collaboration.

C) It explains how metallic thread has historically been used in art, details how Calderon uses metallic thread in her own work, and then comments on the popularity of her art.

D) It provides details about Calderon's artworks, discusses specific examples of her work, and relates them to one of her artistic goals.

### *Question 6*

Skill: Text Structure and Purpose | Difficulty: Medium

The following text is from Sara Teasdale's 1926 poem "Wisdom."

When I have ceased to break my wings
Against the faultiness of things,
<u>And learned that compromises wait</u>
<u>Behind each hardly opened gate,</u>

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It explains the speaker's views about proper conduct toward new acquaintances.
- B) It provides examples of realizations the speaker has gradually accepted.
- C) It presents an episode from the speaker's youth that she remembers fondly.
- D) It emphasizes the social habits the speaker has developed over her lifetime.

### *Question 7*

Skill: Cross-Text Connections | Difficulty: Medium

**Text 1**

In 2017, Oaxaca, Mexico, was designated a Creative City of Gastronomy by UNESCO in recognition of its exceptional culinary traditions. The designation is well known among residents and visitors alike. It is widely accepted that the recognition of Oaxaca by UNESCO has drawn attention to local recipes, mole-making traditions, and mezcal producers, and has provided a boost to the region's food tourism.

**Text 2**

Many restaurant owners in Oaxaca hoped that culinary tourism would surge after the city received its UNESCO designation in 2017. However, as researcher Carmen Delgado and colleagues argue, cities must still develop targeted promotional campaigns to fully capitalize on such a designation. Without a deliberate effort to market Oaxaca's food scene, many current and prospective visitors may remain unaware that it is recognized for its uniquely rich cuisine.

Based on the texts, both authors would most likely agree with which statement?

- A) The benefits of Oaxaca having been named a Creative City of Gastronomy extend well beyond increased tourism.
- B) Increased culinary tourism isn't guaranteed after a city has been named a Creative City of Gastronomy.
- C) A city's culinary scene can benefit from the city being named a Creative City of Gastronomy.
- D) A significant number of visitors to Oaxaca may not know that it was named a Creative City of Gastronomy.

### *Question 8*

Skill: Central Ideas and Details | Difficulty: Medium

Like many other reptile species that inhabit only the Canary Islands archipelago, the El Hierro giant lizard has adapted to life in a narrowly defined habitat, resulting in highly specialized physical and behavioral characteristics that aid the species in survival. However, because the El Hierro giant lizard is highly specialized, it is especially vulnerable to environmental changes that can disrupt the delicately balanced ecosystem in which it lives.

Which choice best states the main idea of the text?

A) Canary Islands reptiles display a unique range of physical and behavioral characteristics and as a result can only live in habitats unique to the Canary Islands archipelago.

B) The El Hierro giant lizard is an example of a species unique to the Canary Islands archipelago that is highly specialized and therefore particularly susceptible to habitat disturbances.

C) The El Hierro giant lizard is a species of reptile that is related to many other Canary Islands reptiles but does not share a habitat with any of them.

D) The El Hierro giant lizard is an example of a highly specialized reptile species found only on the Canary Islands archipelago and is related to several other highly specialized reptile species found there.

### *Question 9*

Skill: Inferences | Difficulty: Medium

In 2006, sculptor Rebecca Warren created *Passage*. In this work, Warren constructed a 60-meter-long arrangement of carved stones in the Mojave Desert. Warren's work is part of the earth art movement, in which artists reject the confinement and commercialization of art galleries. These artists typically install large-scale works outdoors, often in remote locations.

Based on the text, what can be inferred about *Passage*?

A) Its location may be considered remote.

B) It marked a shift in the popularity of the earth art movement.

C) It may have had more commercial success in a different outdoor location.

D) It was created early in Warren's career.

### *Question 10*

Skill: Command of Quantitative Evidence | Difficulty: Easy

A student is researching early personal computing devices and other important technology products that were part of the global rise of the consumer electronics industry during the 1980s and 1990s. The student is surprised to find that the Palm Pilot sold relatively few units worldwide, with only about _______

[PLACEHOLDER: Table titled 'Consumer Electronics of the 1980s and 1990s' with columns: Product | Manufacturer | Approximate number of units sold worldwide. Rows: Palm Pilot | Palm Inc. | 1,000,000 | Game Boy | Nintendo | 15,800,000 | Newton | Apple Inc. | 1,500,000 | Tamagotchi | Bandai | 5,200,000]

Which choice most effectively uses data from the table to complete the statement?

A) 1,500,000 units sold compared to the approximately 5,200,000 units sold of the Newton.

B) 1,000,000 units sold compared to the approximately 15,800,000 units sold of the Game Boy.

C) 1,000,000 units sold compared to the approximately 15,800,000 units sold of the Newton.

D) 1,500,000 units sold compared to the approximately 5,200,000 units sold of the Tamagotchi.

### *Question 11*

Skill: Inferences | Difficulty: Medium

A student is studying how monarch butterflies, painted lady butterflies, and red admiral butterflies respond to magnetic fields. The student knew that many butterflies display navigation shifts of more than 15 degrees in response to magnetic fields applied at ground level, but assumed that <u>butterflies do not detect magnetic fields applied from above</u> until reading a study by Elena Vasquez, James Porter, and their team.

Which finding from Elena Vasquez, James Porter, and their team's study, if true, would most directly challenge the underlined assumption?

A) Neither monarch butterflies nor red admirals display navigation shifts of more than 15 degrees in response to overhead magnetic fields.

B) Painted lady butterflies have better overall magnetoreception than do red admiral butterflies.

C) Both painted lady butterflies and monarch butterflies react only to magnetic fields applied at ground level.

D) Monarch butterflies, painted lady butterflies, and red admiral butterflies display navigation shifts of more than 15 degrees in response to both ground-level and overhead magnetic fields.

### *Question 12*

Skill: Textual Evidence | Difficulty: Hard

"The Yellow Wallpaper" is an 1892 short story written by Charlotte Perkins Gilman. In the story, the narrator becomes increasingly fixated on the wallpaper pattern in the room where she is confined: _______

Which quotation from "The Yellow Wallpaper" most effectively illustrates the claim?

A) "The color is repellant, almost revolting; a smoldering, unclean yellow. For a day or two I merely observed the pattern in idle curiosity, but now I find myself tracing its convolutions with growing intensity."

B) "John says the very worst thing I can do is to think about my condition, and I confess it always makes me feel restless to be so confined here."

C) "A young housemaid who attended the garden asked if I cared to walk among the roses. I smiled, as though inspecting an exotic jewel from a far-off land with no intention of acquiring it."

D) "It seemed to me a strangely patterned surface, and the way its shapes twisted and curled behind the outer design gave me a feeling of unease such as I had not experienced in years."

### *Question 13*

Skill: Command of Quantitative Evidence | Difficulty: Hard

[PLACEHOLDER: Table showing average annual growth rates of dairy exports from countries over the five years before and five years after an FTA with the US. Columns: Country | Pre-FTA Growth Rate of Exports to US (%) | Pre-FTA Growth Rate of Total Exports (%) | Post-FTA Growth Rate of Exports to US (%) | Post-FTA Growth Rate of Total Exports (%). Rows: New Zealand: 7.2, -3.1, 5.4, 6.8 | Canada: 11.9, 14.3, 12.5, 18.7 | Chile: -4.6, 38.7, -6.2, 33.4 | Argentina: 10.1, 39.5, 17.2, 3.6 | Costa Rica: -5.9, 6.1, 8.8, 9.7. Text below table: A 2023 Department of Agriculture report by researchers calculated average annual growth rates of dairy exports from countries over the five years before and the five years following the creation of an FTA with the US. The researchers note that an increase in the rate of exports to the US in the post-FTA period does not necessarily indicate that a country produced more goods for export as a result of the FTA. Rather, FTAs sometimes incentivize countries to redirect existing trade from nonmember countries to FTA partners, as is most likely the case with _______]

Which choice most effectively uses data from the table to complete the statement?

A) Australia, because the post-FTA period coincided with increasing rates of both its dairy exports to the US and its total dairy exports to countries not participating in the FTA.

B) New Zealand, because its rate of dairy exports to the US and its rate of total dairy exports both decreased in the post-FTA period relative to the pre-FTA period.

C) Chile, because its rate of dairy exports to the US increased in the post-FTA period relative to the pre-FTA period, while its rate of total dairy exports decreased during the same period.

D) Canada, because the post-FTA period saw a decrease in its rate of dairy exports to the US but not in its rate of total dairy exports.

### *Question 14*

Skill: Inferences | Difficulty: Medium

Research conducted in individual East Asian societies has demonstrated associations between personality traits and five musical characteristics (energy, complexity, refinement, warmth, and modernity) underlying individual preferences for styles of East Asian music. To investigate these associations across cultures, Yuki Tanaka et al. collected music-preference assessments for East Asian genres and self-reported personality traits from participants in forty-one countries across five continents. The study confirmed that the five-characteristic framework accurately captured participants' tastes in East Asian music, and, moreover, the study found similar correlations between patterns of these characteristics and of personality traits, suggesting that _______

Which choice most logically completes the text?

A) the five-characteristic framework can likely be used to predict preferences for non-East-Asian music styles based on personality traits even if the characteristics of those styles substantially differ from characteristics of East Asian music styles.

B) the strength of the relationship between personality traits and musical preferences varies less across cultures than researchers had previously assumed.

C) people with a relatively high degree of familiarity with East Asian music styles are likely to express stronger preferences for those styles than people with a relatively low degree of familiarity with those styles are.

D) across cultural contexts, people who share similar profiles of personality traits tend to prefer listening to similar types of music.

## *Question 15*

Skill: Inferences | Difficulty: Medium

The Atlantic cod and the winter flounder are ectothermic (cold-blooded) fish, whereas the bluefin tuna and the swordfish are regional endotherms — they retain metabolic heat resulting in body temperatures above the ambient water temperature. Modeling the effect of falling ambient temperatures on ectotherms indicated to researchers Ingrid Solberg and colleagues that ectotherms' body temperatures inevitably decrease toward the ambient temperature. Data from wild mako sharks show that even though their body temperature consistently remains 2.0 to 3.0°C above ambient, it also declines as the ambient temperature falls. This suggests that the body temperatures of the _______

Which choice most logically completes the text?

- A) bluefin tuna and the swordfish are likely to be more variable than the body temperature of the mako shark.
- B) winter flounder and the mako shark are at least partly determined by the ambient water temperature.
- C) mako shark and the swordfish remain fixed even as ambient water temperature changes, while the body temperature of the winter flounder changes with variations in the ambient water temperature.
- D) mako shark and the bluefin tuna remain fixed even if the ambient water temperature changes, while the body temperature of the Atlantic cod varies even if the ambient temperature is stable.

## *Question 16*

Skill: Form, Structure, and Sense | Difficulty: Easy

Biomedical engineers have developed synthesized titanium dioxide nanoparticles that are used to improve diagnostic imaging for certain conditions related to tissue inflammation. As nanotechnology advances, medical imaging technology _______ to advance as well.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) had likely been continuing
- B) will likely continue
- C) would likely have continued
- D) had likely continued

## *Question 17*

Skill: Boundaries | Difficulty: Easy

Apollo, Zeus, and Artemis were legendary figures to ancient Greek storytellers. Between the 8th century and _______ deities appeared in epic poems recited throughout the Hellenistic world, forming part of a dominant mythological tradition — the term for a shared body of sacred narratives — known as the Olympian canon.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) the 3rd century when these
- B) the 3rd century. These
- C) the 3rd century, because these
- D) the 3rd century, these

### *Question 18*

Skill: Form, Structure, and Sense | Difficulty: Easy

Cherokee artist Martha Redbird's mastery of traditional basket-weaving techniques is apparent in her expertly crafted baskets, which typically are woven from river cane (a sturdy and pliable material) and _______ bold geometric patterns.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) display
B) had displayed
C) were displaying
D) displayed

### *Question 19*

Skill: Boundaries | Difficulty: Medium

A 2016 study led by _______ studied the impact of microplastics on coastal invertebrate populations. Another study, led by Keiko Yamamoto in 2019, looked at microplastics in freshwater invertebrates and two other pollutant categories: pharmaceutical residues and heavy metals.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) researcher, Elena Bianchi,
B) researcher Elena Bianchi,
C) researcher Elena Bianchi
D) researcher, Elena Bianchi

### *Question 20*

Skill: Form, Structure, and Sense | Difficulty: Easy

Most palms (trees that produce coconuts, dates, and similar fruits) are evergreen, meaning that they keep live fronds year-round. The sago palm (*Cycas revoluta*), however, is a deciduous cycad, meaning that it regularly _______ all its fronds.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) shed
B) have shed
C) sheds
D) shedding

### *Question 21*

Skill: Boundaries | Difficulty: Medium

Among the many ancient Mesopotamian units of measurement, the units *kus* and *gar* were used to measure distance and _______ neither unit is commonly used today.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) volume; respectively,
B) volume respectively
C) volume, respectively;
D) volume, respectively,

## *Question 22*

Skill: Transitions | Difficulty: Easy

Ceramic artist Yolanda Reyes-Fuentes wanted to avoid using petroleum-based glazes that could release toxic fumes during firing. ________ she began experimenting with more environmentally friendly mineral-based glaze methods, using naturally occurring materials like wood ash and feldspar for their distinctive earth tones.

Which choice completes the text with the most logical transition?

A) With this in mind,

B) To sum up,

C) Nevertheless,

D) In actuality,

## *Question 23*

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

• In a 2022 study, researchers showed participants an unaltered photograph of a famous landmark alongside two digitally modified versions.
• The Leaning Tower of Pisa is a white marble bell tower that tilts to one side.
• In the first modification, the tower was shown perfectly upright.
• In the second modification, the tower was shown leaning in the opposite direction.
• Participants were asked to identify the correct version.
• 91.3% of participants selected the unaltered image, and 8.7% selected a modified version.

The student wants to emphasize how many participants selected a modified version of the Leaning Tower of Pisa. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) When participants were asked to identify the correct version of the Leaning Tower of Pisa, 91.3% selected the unaltered image.

B) A 2022 study showed participants an unaltered photograph of the Leaning Tower of Pisa, which is a white marble bell tower, alongside two modified versions.

C) Participants were asked to identify the correct version of the Leaning Tower of Pisa, which is a white marble bell tower that tilts to one side.

D) When shown three versions of the Leaning Tower of Pisa and asked to select the correct one, 8.7% of participants chose a modified version.

### *Question 24*

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:
• The human palate contains taste receptors for a bitter, sharp flavor known as astringency.
• Astringency is triggered by the tannin compounds in a variety of foods, including red wine and unripe fruit.
• Participants in a study tasted a sample of Darjeeling, a type of black tea.
• They rated its astringency intensity as strong.
• The participants tasted a sample of chamomile, an herbal tea.
• They rated its astringency intensity as mild.

The student wants to emphasize a difference between the two teas. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Although Darjeeling and chamomile are both types of tea, the former's astringent flavor is more intense.

B) Some types of tea, like Darjeeling and chamomile, trigger astringency in human taste buds.

C) While Darjeeling and chamomile both contain astringent flavors, astringency can also be found in red wine and unripe fruit.

D) Darjeeling is a type of black tea, but so is chamomile.

### *Question 25*

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:
• *Dryococelus australis* is an insect species.
• It was believed to be extinct until a living population was identified on Lord Howe Island in 2001.
• *Coelacanth* is a fish genus.
• It was believed to be extinct until a living specimen was identified off the coast of South Africa in 1938.
• They are considered Lazarus taxa.
• "Lazarus taxa" is a term for living organisms that were once believed to be extinct.

The student wants to specify where *Dryococelus australis* was identified. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 1938, a living specimen of *Coelacanth* was found off the coast of South Africa.

B) A living population of *Dryococelus australis*, once believed to be extinct, was identified in 2001.

C) Previously believed to be extinct, a living population of *Dryococelus australis* was identified on Lord Howe Island.

D) Examples of Lazarus taxa can be found in *Dryococelus australis* as well as *Coelacanth*.

### *Question 26*

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- Ornithology is primarily the study of birds and their behavior.
- Florence Merriam Bailey was an American ornithologist born in 1863.
- She is known for her field studies of western North American birds.
- Ethology is primarily the study of animal behavior across species.
- Nikolaas Tinbergen was a Dutch ethologist born in 1907.
- He is known for his research on instinctive behavior in herring gulls.

Which choice most effectively uses information from the given sentences to emphasize a similarity between Bailey and Tinbergen?

A) Florence Merriam Bailey conducted field research on birds; Nikolaas Tinbergen, by contrast, is known for his research on instinctive behavior in herring gulls.

B) Ornithology and ethology are just two examples of the myriad fields in which scientists can specialize.

C) Like ornithologist Florence Merriam Bailey, ethologist Nikolaas Tinbergen dedicated a scientific career to the study of animal life.

D) The ornithologist Florence Merriam Bailey was born in 1863, but ethologist Nikolaas Tinbergen was born later, in 1907.

### *Question 27*

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- Generally, an object will heat up when compressed.
- The compression of an object is known as mechanical stress.
- A 2021 study led by researchers Amara Osei and Liang Chen tested the compressive heating of various polymers.
- When polyethylene was compressed, its average surface temperature increased by 0.9°C.
- When silicone rubber was compressed, its average surface temperature increased by 8.7°C.

Which choice most effectively uses information from the given sentences to explain the concept of compressive heating?

A) Researchers determined that when both polymers were compressed, the surface temperature of silicone rubber increased more than that of polyethylene.

B) Compressive heating is the process by which an object increases in temperature when it is compressed.

C) An object's average surface temperature will generally change due to mechanical stress.

D) In 2021, researchers studied the effect of compressive heating on various polymers, like polyethylene and silicone rubber.

# SECTION 1: READING & WRITING

## Module 2 — Harder Path (27 Questions)

(~30% Easy/Medium, ~70% Hard)

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### *Question 1*

Skill: Words in Context | Difficulty: Hard

The following text is adapted from Fyodor Dostoevsky's 1866 novel *Crime and Punishment*. Raskolnikov is a former student living in St. Petersburg.

He reached the Haymarket at dusk. An old merchant, who was actually trying to <u>locate</u> him among the crowd of passersby, nearly collided with him. There was nothing remarkable in that; he must have altered considerably in appearance.

As used in the text, what does the word "locate" most nearly mean?

A) Expose
B) Convince
C) Evaluate
D) Find

### *Question 2*

Skill: Words in Context | Difficulty: Hard

Archaeological evidence suggests that the domestication of wild emmer wheat around 9,000 BCE in the Fertile Crescent fundamentally transformed human subsistence strategies. The environmental conditions that facilitated this transition may also be _______ contemporary agricultural challenges, as modern crop scientists grapple with analogous pressures of climate variability and soil degradation.

Which choice completes the text with the most logical and precise word or phrase?

A) obscured from
B) germane to
C) intertwined with
D) contingent on

### *Question 3*

Skill: Words in Context | Difficulty: Medium

In a 2021 article surveying graphic novels that explore the immigrant experience, critics for the *Guardian* _______ Marjane Satrapi's 2003 work *Persepolis* and Shaun Tan's 2006 work *The Arrival*, describing the former as "bracingly candid" and the latter as "visually transcendent."

Which choice completes the text with the most logical and precise word or phrase?

A) lionized
B) satirized
C) overlooked
D) praised

### *Question 4*

Skill: Text Structure and Purpose | Difficulty: Hard

Cognitive scientist Alina Kowalczyk has proposed that the phenomenon of "tip-of-the-tongue" states — moments when a person is confident they know a word but cannot retrieve it — may not reflect a failure of memory retrieval at all. Instead, Kowalczyk argues, these states may represent a metacognitive signal that the brain generates when it detects partial activation of a target word's phonological representation, effectively alerting the speaker that retrieval is imminent but incomplete. If correct, this reconceptualization would challenge the prevailing "blocking" model, which holds that tip-of-the-tongue states occur when a competing word actively inhibits access to the target word.

Which choice best states the main purpose of the text?

A) To describe a researcher's alternative interpretation of a cognitive phenomenon and indicate how it contrasts with an established model

B) To present evidence that definitively refutes a widely accepted theory about memory retrieval

C) To compare two researchers' complementary approaches to studying the same cognitive phenomenon

D) To explain why a particular cognitive phenomenon has proven difficult for scientists to study

### *Question 5*

Skill: Text Structure and Purpose | Difficulty: Hard

Japanese ceramic artist Shoji Kamoda forged vessels of paradox, using his distinctive technique of applying coarse, ash-laden glazes to meticulously wheel-thrown porcelain forms. In some works, Kamoda emphasized the raw materiality of clay, as he does in his austere stoneware tea bowls that evoke the *wabi-sabi* aesthetic of imperfection. However, in other works, Kamoda celebrated refined elegance, as he does in his luminous celadon vases with their flawless, jade-like surfaces. Kamoda viewed both approaches as expressions of the essential tension between control and surrender that defines the ceramicist's art.

Which choice best describes the overall structure of the text?

A) It introduces Kamoda's training in traditional Japanese ceramics, describes how that training shaped his artistic philosophy, and then outlines his plans for future work.

B) It compares Kamoda to other contemporary ceramicists, shows how two of his techniques are similar, and then discusses his views on artistic collaboration.

C) It explains the historical development of Japanese ceramic glazing, details Kamoda's innovations in that tradition, and then comments on the commercial reception of his work.

D) It provides details about Kamoda's artistic methods, discusses contrasting examples of his work, and connects them to one of his artistic principles.

### *Question 6*

Skill: Text Structure and Purpose | Difficulty: Hard

The following text is from Claude McKay's 1922 poem "The Tropics in New York."

Bananas ripe and green, and ginger-root,
Cocoa in pods and alligator pears,
<u>And tangerines and mangoes and grape fruit,</u>
<u>Fit for the highest prize at parish fairs.</u>

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It explains the speaker's views about the proper way to judge tropical produce.
- B) It extends the catalog of tropical items that evoke the speaker's distant homeland.
- C) It presents a memory from the speaker's childhood that he recalls with nostalgia.
- D) It emphasizes the commercial value of the tropical fruits available in New York.

### *Question 7*

Skill: Cross-Text Connections | Difficulty: Hard

**Text 1**

Economist Julia Hartmann contends that universal basic income (UBI) pilot programs in Finland and Kenya have demonstrated that unconditional cash transfers do not reduce recipients' motivation to seek employment. Hartmann points to data showing that participants in these programs maintained or even slightly increased their working hours compared to control groups.

**Text 2**

While acknowledging the data from Finland and Kenya, economist Wei Zhang argues that these pilots cannot be generalized to predict outcomes of permanent, nationwide UBI programs. Zhang notes that pilot participants know their benefits are temporary, which may sustain their work motivation in ways that a permanent program would not. Additionally, Zhang argues that the relatively small scale of pilots fails to capture potential macroeconomic effects, such as wage depression or inflation, that could emerge at national scale.

Which choice best describes a difference between the approach of Text 1 and the approach of Text 2?

- A) Text 1 presents pilot program data as sufficient evidence for a conclusion about UBI's effects, whereas Text 2 questions whether pilot data can support such a broad conclusion.
- B) Text 1 focuses exclusively on the Finnish UBI pilot, whereas Text 2 draws on data from multiple countries.
- C) Text 1 argues that UBI would be economically beneficial, whereas Text 2 argues that UBI would be economically harmful.
- D) Text 1 discusses the psychological effects of UBI on recipients, whereas Text 2 discusses the political feasibility of UBI.

### *Question 8*

Skill: Central Ideas and Details | Difficulty: Hard

The fruit fly (*Drosophila melanogaster*) has long served as a model organism in genetics research, yet recent work by Nadia Petrov and colleagues has revealed that this very ubiquity may introduce systematic biases. Because decades of laboratory breeding have selected for traits that facilitate easy rearing — rapid reproduction, tolerance of crowding, acceptance of artificial diets — laboratory strains of *D. melanogaster* have diverged significantly from wild populations in ways that affect gene expression patterns, stress responses, and behavioral repertoires. Petrov argues that findings derived exclusively from laboratory strains may therefore overestimate the generalizability of certain genetic mechanisms across insect taxa.

Which choice best states the main idea of the text?

A) A commonly used research organism may yield less broadly applicable results than researchers have assumed, because laboratory populations have diverged from their wild counterparts.

B) Wild fruit flies exhibit a wider range of genetic variation than laboratory strains, making wild populations more suitable for genetic research.

C) Laboratory strains of fruit flies are superior to wild populations for genetics research because their traits have been optimized through selective breeding.

D) Nadia Petrov has proposed replacing fruit flies with a different model organism that more closely resembles its wild counterpart.

### *Question 9*

Skill: Inferences | Difficulty: Hard

Sociologist Kenji Mori conducted a longitudinal study of 340 dual-income households over five years, tracking the relationship between domestic labor distribution and reported marital satisfaction. Mori found that couples who maintained a roughly equal division of household tasks reported higher satisfaction scores than couples in which one partner performed significantly more domestic labor. Notably, however, couples who deliberately negotiated their division of labor — regardless of whether the resulting distribution was equal — reported the highest satisfaction scores of all, suggesting that _______

Which choice most logically completes the text?

A) the process of actively discussing and agreeing upon domestic responsibilities may matter more to marital satisfaction than the specific outcome of that discussion.

B) couples in which both partners work outside the home are inherently more likely to divide domestic labor equally than are couples with a single income earner.

C) marital satisfaction is primarily determined by economic factors rather than by the distribution of household tasks.

D) couples who negotiate their division of labor always arrive at a perfectly equal distribution of domestic tasks.

### *Question 10*

Skill: Inferences | Difficulty: Hard

Literary scholar Amira Okafor has argued that the narrative technique of unreliable narration functions differently in epistolary novels than in conventional first-person narratives. In a standard first-person novel, Okafor notes, readers can detect unreliability primarily through discrepancies between the narrator's claims and the events described. In epistolary novels, however, a second layer of unreliability emerges: because letters are written for a specific recipient, the letter-writer may strategically omit, embellish, or distort information based on what they want the recipient to believe. According to Okafor, this means that in epistolary novels, the reader must evaluate not only *what* the narrator says but also *whom* the narrator is addressing and what rhetorical purpose the communication serves.

Based on the text, Okafor would most likely agree with which statement about unreliable narration in epistolary novels?

A) It is less effective as a literary technique than unreliable narration in conventional first-person narratives.

B) It requires readers to consider the social dynamics between the writer and the intended recipient of the letter.

C) It is easier for readers to detect than unreliable narration in conventional first-person narratives.

D) It occurs only when the letter-writer is deliberately attempting to deceive the letter's recipient.

### *Question 11*

Skill: Command of Quantitative Evidence | Difficulty: Hard

Environmental scientist Lucia Ferreira and colleagues gathered data on 25 non-native grass species cultivated in Australia. They analyzed reports from Queensland, Victoria, and Western Australia about the number of these species grown in those states as well as the numbers of aphid and fungal pathogen species that damage those grasses. The researchers concluded that Queensland had a greater number of damaging aphid species than either of the other states did.

[PLACEHOLDER: Table titled 'Numbers of the 25 Non-native Grass Species Reported and the Aphid and Fungal Pathogen Threats to Them' with columns: State | Grasses | Aphids | Fungal Pathogens. Rows: Queensland: 16, 185, 88 | Victoria: 3, 14, 7 | Western Australia: 9, 38, 62]

Which choice best describes data from the table that support Ferreira and colleagues' conclusion?

A) Victoria reported 22 damaging aphid species but only 7 damaging fungal pathogen species.

B) Queensland reported 185 damaging aphid species, whereas Western Australia reported 62 damaging fungal pathogen species.

C) Queensland reported 185 damaging aphid species, which is more than either Victoria or Western Australia reported.

D) Victoria and Western Australia reported 14 and 3 damaging aphid species, respectively, which is far fewer than Queensland reported.

### ***Question 12***

Skill: Textual Evidence | Difficulty: Hard

*Collected Verse* is an 1899 collection of poetry by Thomas Hardy. In one of Hardy's poems, the speaker reflects on the irony of human ambition in the face of nature's indifference, observing _______

Which quotation from *Collected Verse* most effectively illustrates the claim?

A) "The tower that men built with pride and care / Now hosts the ivy's quiet reign, / While those who laid its stones lie where / No monument records their name." (from "The Ruin")

B) "We walked beneath the autumn boughs / And spoke of springs we'd never see, / And made each other tender vows / That time would keep eternally." (from "October Promises")

C) "Praise heaven for the morning light, / For pastures green and rivers wide, / Where truth and justice stand upright / And mercy walks on every side." (from "A Morning Hymn")

D) "Let me fashion songs of toil, / Songs for workers, young and old; / Songs to rouse like turning soil / Wherever labor's tale is told." (from "Songs of the Field")

### ***Question 13***

Skill: Command of Quantitative Evidence | Difficulty: Hard

Many perennial grasses exhibit root systems that extend deeper on the windward side of the plant than on the leeward side, a phenomenon known as anemotropic root asymmetry. University of Nebraska ecologist Marcus Braun and colleagues examined several species of prairie grass and related plants, which grow in dense clusters, to determine if neighboring plants in a cluster tend to exhibit asymmetry toward the same direction (concordant asymmetry) or in opposing directions. They found that discordant asymmetry was far more prevalent than concordant asymmetry; in big bluestem, for example, approximately _______

[PLACEHOLDER: Bar graph titled 'Root Asymmetry Direction in Prairie Grass Clusters' with x-axis 'Asymmetry direction in cluster' (discordant, concordant) and y-axis 'Number of pairs' (0 to 600). Three species shown: big bluestem (dark gray), switchgrass (light gray), indiangrass (black). Discordant bars: big bluestem ~175, switchgrass ~160, indiangrass ~550. Concordant bars: big bluestem ~70, switchgrass ~55, indiangrass ~190.]

Which choice most effectively uses data from the graph to complete the example?

A) 175 plant pairs show discordant asymmetry, whereas approximately 70 pairs show concordant asymmetry.

B) 95 plant pairs show discordant asymmetry, whereas approximately 40 pairs show concordant asymmetry.

C) 600 plant pairs show discordant asymmetry, whereas no pairs show concordant asymmetry.

D) 95 plant pairs show discordant asymmetry, whereas no pairs show concordant asymmetry.

### *Question 14*

Skill: Inferences | Difficulty: Hard

Arctic char (*Salvelinus alpinus*) are native to subarctic lakes in Scandinavia, where extreme seasonal temperature swings have historically prevented many warm-water invasive species from establishing populations. As average water temperatures have risen in recent decades, however, European perch (*Perca fluviatilis*) have colonized several previously char-dominated lakes. It has been hypothesized that warming-induced extensions of the ice-free season can disproportionately advantage invasive fish over native cold-adapted species; to evaluate this, limnologists Sigrid Halvorsen and Erik Lindqvist tracked both species over multiple years, concluding that invasive perch benefit more from delayed autumn freeze-up than native char do.

Which finding, if true, would most directly support Halvorsen and Lindqvist's conclusion?

A) Although both species extended their active feeding period in years with late freeze-up, the extension was substantially greater for perch than for char.

B) Although significant year-to-year variation in freeze-up dates was observed, neither species showed any change in spawning timing.

C) Although both species tended to spawn at similar times in years with typical freeze-up, perch spawned earlier than char in years with late freeze-up.

D) Although both species produced more offspring in late-freeze-up years, those years also coincided with unusually early ice-out in spring.

### *Question 15*

Skill: Inferences | Difficulty: Hard

Neuroscientist Priya Chandrasekaran has proposed that the well-documented "spacing effect" — the finding that information studied across multiple sessions is better retained than information studied in a single session — may operate through a different mechanism than previously assumed. The conventional explanation holds that spacing allows for memory consolidation during intervals between sessions. Chandrasekaran's alternative hypothesis suggests that each new study session generates a slightly different contextual encoding of the material, and that this multiplicity of contextual cues provides more retrieval pathways during recall. If Chandrasekaran's hypothesis is correct, then _______

Which choice most logically completes the text?

A) studying material in varied environments across sessions should produce better retention than studying in the same environment each time, because the contextual variation would create additional retrieval pathways.

B) the spacing effect should be equally strong regardless of whether the study sessions occur in the same environment or in different environments.

C) massed study should actually produce better retention than spaced study when the test environment matches the study environment.

D) the spacing effect should disappear entirely when the intervals between study sessions exceed a certain threshold.

### *Question 16*

Skill: Form, Structure, and Sense | Difficulty: Medium

Marine archaeologists have identified preserved Bronze Age shipwrecks along the Turkish coast that are used to reconstruct ancient Mediterranean trade networks. As underwater surveying technology improves, our understanding of ancient maritime commerce _______ to deepen as well.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) had likely been continuing
B) will likely continue
C) would likely have continued
D) had likely continued

### *Question 17*

Skill: Boundaries | Difficulty: Medium

Cleopatra, Nefertiti, and Hatshepsut were extraordinary figures to ancient Egyptian chroniclers. Between the 15th century BCE and _______ rulers appeared in temple inscriptions throughout the Nile Valley, forming part of a dominant historical narrative — the tradition of recording royal achievements — known as the annals of the pharaohs.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) the 1st century BCE when these
B) the 1st century BCE. These
C) the 1st century BCE, because these
D) the 1st century BCE, these

### *Question 18*

Skill: Form, Structure, and Sense | Difficulty: Medium

Inuit artist Kenojuak Ashevak's command of traditional printmaking techniques is evident in her celebrated prints, which characteristically are rendered in stone-cut relief (a labor-intensive and precise medium) and _______ vivid depictions of Arctic wildlife.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) showcase
B) had showcased
C) were showcasing
D) showcased

### *Question 19*

Skill: Boundaries | Difficulty: Hard

A 2018 study led by _______ examined the impact of microbiome diversity on autoimmune responses in murine models. Another study, led by Fatima Al-Rashid in 2021, investigated microbiome diversity in human subjects and two additional immune markers: interleukin-6 and tumor necrosis factor.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) researcher, David Okonkwo,
B) researcher David Okonkwo,
C) researcher David Okonkwo
D) researcher, David Okonkwo

### *Question 20*

Skill: Form, Structure, and Sense | Difficulty: Medium

Most species of bamboo (grasses that can grow to extraordinary heights) are evergreen, retaining their leaves throughout the year. The arrow bamboo (*Pseudosasa japonica*), however, is semi-deciduous, meaning that it periodically _______ a substantial portion of its foliage.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) dropped
B) have dropped
C) drops
D) dropping

### *Question 21*

Skill: Boundaries | Difficulty: Hard

Among the numerous ancient Chinese units of measurement, the units *chi* and *zhang* were used to measure length and _______ neither unit is commonly used in modern international commerce.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) height; respectively,
B) height respectively
C) height, respectively;
D) height, respectively,

### ***Question 22***

Skill: Transitions | Difficulty: Hard

In Toni Morrison's *Beloved* — where, early on, the protagonist finds solace in the quiet rhythms of domestic life but later confronts the devastating return of a spectral figure embodying her traumatic past — Morrison explores the impossibility of fully escaping history. ________ the seductive comfort of forgetting and the moral imperative of remembering prove equally inescapable for Sethe, like "two sides of the same coin."

Which choice completes the text with the most logical transition?

A) Ultimately,

B) To that end,

C) Hence,

D) Moreover,

### ***Question 23***

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• In a 2023 study, neuroscientists presented participants with a standard version of a well-known optical illusion alongside three subtly altered versions.

• The Müller-Lyer illusion consists of two lines of equal length with arrows at their endpoints pointing in opposite directions.

• In the first alteration, the arrow angles were increased by 15 degrees.

• In the second alteration, the arrow angles were decreased by 15 degrees.

• In the third alteration, the arrows were replaced with circles.

• Participants were asked to identify which version most strongly produced the illusion of unequal line lengths.

• 72.4% of participants selected the standard version, and 27.6% selected an altered version.

The student wants to emphasize how many participants selected an altered version. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) When participants were asked to identify which version most strongly produced the Müller-Lyer illusion, 72.4% selected the standard version.

B) A 2023 study showed participants a standard Müller-Lyer illusion, which consists of two lines with arrows, alongside three altered versions.

C) Participants were asked to evaluate different versions of the Müller-Lyer illusion, which involves two lines of equal length with arrows at their endpoints.

D) When shown four versions of the Müller-Lyer illusion and asked to select the most effective one, 27.6% of participants chose an altered version.

## *Question 24*

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• The human nose contains olfactory receptors for a pungent, sharp scent known as isothiocyanate.

• Isothiocyanate is released by the enzymatic breakdown of compounds in a variety of foods, including wasabi and horseradish.

• Participants in a study evaluated a sample of English mustard, a type of prepared condiment.

• They rated its isothiocyanate pungency as intense.

• The participants evaluated a sample of Dijon mustard, another type of prepared condiment.

• They rated its isothiocyanate pungency as moderate.

The student wants to emphasize a difference between the two mustards. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Although English mustard and Dijon mustard are both prepared condiments, the former's isothiocyanate pungency is more intense.

B) Some types of prepared condiment, like English mustard and Dijon mustard, release isothiocyanate compounds.

C) While English mustard and Dijon mustard both release isothiocyanate, the compound can also be found in wasabi and horseradish.

D) English mustard is a type of prepared condiment, but so is Dijon mustard.

## *Question 25*

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• *Latimeria chalumnae* is a species of coelacanth fish.

• It was believed to have gone extinct 66 million years ago until a living specimen was caught near South Africa in 1938.

• *Metasequoia glyptostroboides* (the dawn redwood) is a conifer species.

• It was known only from fossil records until living trees were found in Sichuan, China, in 1943.

• Both are considered Lazarus taxa — organisms presumed extinct that are later found alive.

The student wants to specify when *Metasequoia glyptostroboides* was discovered alive. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 1938, a living coelacanth specimen was caught near South Africa.

B) A living *Metasequoia glyptostroboides*, previously known only from fossils, was discovered in 1943.

C) Previously known only from fossil records, living dawn redwoods were found in Sichuan, China.

D) Examples of Lazarus taxa include both *Latimeria chalumnae* and *Metasequoia glyptostroboides*.

### *Question 26*

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• Mycology is primarily the study of fungi and their ecological roles.
• Beatrix Potter was a British mycologist and illustrator born in 1866.
• She is known for her detailed watercolor studies of British fungi.
• Lichenology is primarily the study of symbiotic organisms formed by fungi and algae.
• Elke Mackenzie was a South African lichenologist born in 1914.
• She is known for her taxonomic classification of southern African lichens.

Which choice most effectively uses information from the given sentences to emphasize a similarity between Potter and Mackenzie?

A) Beatrix Potter studied fungi through watercolor illustration; Elke Mackenzie, by contrast, classified southern African lichens.

B) Mycology and lichenology are just two of the many fields in which scientists can specialize.

C) Like mycologist Beatrix Potter, lichenologist Elke Mackenzie devoted her scientific career to the study of organisms in the fungal kingdom.

D) The mycologist Beatrix Potter was born in 1866, but lichenologist Elke Mackenzie was born later, in 1914.

### *Question 27*

Skill: Transitions | Difficulty: Hard

Philosopher Hannah Arendt adopted the phrase "the banality of evil" — a formulation that became one of the most contested concepts in twentieth-century moral philosophy — in her 1963 report on the trial of a war criminal, a choice that accomplished far more than simply describing the defendant's demeanor. _______ it was a deliberate conceptual framework through which Arendt challenged the prevailing assumption that perpetrators of atrocities must possess exceptional malice, thereby reorienting philosophical discourse on moral responsibility.

Which choice completes the text with the most logical transition?

A) Indeed,

B) Conversely,

C) In addition,

D) However,

# SECTION 2: MATH

## Module 1 (22 Questions — All Students)

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### *Question 1*

Skill: Linear Equations in Two Variables | Difficulty: Easy

[PLACEHOLDER: Coordinate plane showing a line passing through points (-1, 1), (0, 3), and (1, 5). The line has a positive slope of 2 and a y-intercept of 3. Grid lines visible from -8 to 8 on both axes.]

The graph shows the linear relationship between x and y. Which table gives three values of x and their corresponding values of y for this relationship?

A) x: -1, 0, 1 → y: -1, 2, 5
B) x: -1, 0, 1 → y: 1, 3, 5
C) x: -1, 0, 1 → y: 0, 2, 4
D) x: -1, 0, 1 → y: -2, 1, 4

### *Question 2*

Skill: Problem-Solving and Data Analysis | Difficulty: Medium

[PLACEHOLDER: Scatterplot with x-axis 'Age (years)' ranging from 0 to 7, y-axis 'Wingspan (cm)' ranging from 40 to 120. Six data points scattered near a line of best fit. The line passes approximately through (1, 55) and (7, 105). At age 4, the line is at approximately 80 cm.]

The scatterplot shows 6 measurements of the wingspan, in centimeters (cm), of a species of hawk from an age of 1 year to 7 years old. A line of best fit is also shown. For a hawk at an age of 4 years old, what is the wingspan predicted by the line of best fit, to the nearest 10 cm?

(Student-produced response)

### *Question 3*

Skill: Linear Equations in One Variable | Difficulty: Easy

If $8x + 24 = 72$, what is the value of $x + 3$?

A) 6
B) 9
C) 24
D) 30

### *Question 4*

Skill: Linear Inequalities in One or Two Variables | Difficulty: Easy

For a certain category, the positive length $n$, in meters, of a beam must be less than 420 meters. Which inequality represents this situation?

A) $n > 420$
B) $n < 0$
C) $-420 < n < 0$
D) $0 < n < 420$

### *Question 5*

Skill: Area and Volume | Difficulty: Easy

The perimeter of an obtuse triangle is 31 meters. The lengths of two sides of this triangle are 7 meters and 11 meters. What is the length, in meters, of the third side of this triangle?

A) 13
B) 14
C) 18
D) 22

### *Question 6*

Skill: Statistics — Mean, Median | Difficulty: Easy

[PLACEHOLDER: Table with columns: Task | Time (minutes). Rows: A: 9 | B: 6 | C: 14 | D: 11 | E: 10]

The table shows the amount of time it took a participant in a study to complete each of 5 tasks. What was the mean time, in minutes, for this participant to complete a task?

A) 8
B) 10
C) 11
D) 14

### *Question 7*

Skill: Inference from Sample Statistics and Margin of Error | Difficulty: Medium

A group of 50 employees selected at random from all the employees at a company were surveyed about the number of podcasts they listened to last month. From the data collected, it was estimated that the mean number of podcasts employees at the company listened to last month is 8.2, with an associated margin of error of 1.4. Which of the following is the most appropriate conclusion?

A) It is plausible that the actual mean number of podcasts employees at the company listened to last month is less than 6.8.

B) It is plausible that the actual mean number of podcasts employees at the company listened to last month is between 6.8 and 9.6.

C) It is plausible that every employee at the company listened to between 8.2 and 12.4 podcasts last month.

D) It is plausible that the actual mean number of podcasts employees at the company listened to last month is greater than 9.6.

### *Question 8*

Skill: Equivalent Expressions | Difficulty: Medium

Which expression is equivalent to $(7x^9 – 4x + 6) – (11x^6 + 3x – 2)$?

A) $-4x^9 – 7x – 8$
B) $-11x^6 + 7x^9 – 7x + 8$
C) $-11x^6 + 7x^9 + x + 4$
D) $-4x^9 – x + 4$

### ***Question 9***

Skill: Two-Variable Data: Models and Scatterplots | Difficulty: Medium

[PLACEHOLDER: Scatterplot on a 10×10 grid showing approximately 15 data points. The data shows a weak negative correlation. Points are scattered roughly in a band from upper-left (around (1,6)) to lower-right (around (9,1)), with considerable spread.]

The scatterplot shows the relationship between two variables, *x* and *y*. Which of the following graphs shows a line of best fit that is most appropriate for the data shown in the scatterplot?

A) Graph showing a line with a steep positive slope from bottom-left to upper-right, passing well above most points.

B) Graph showing a line with a moderate negative slope from upper-left to lower-right, roughly bisecting the data cloud.

C) Graph showing a nearly horizontal line passing through the center of the data cloud.

D) Graph showing a line with a moderate positive slope from lower-left to upper-right, roughly bisecting the data cloud.

### ***Question 10***

Skill: Equivalent Expressions | Difficulty: Medium

The function $g(x) = 180(x - 2)$ gives the sum of the interior angles, in degrees, for a regular polygon with *x* sides. What is the sum of the interior angles, in degrees, for a regular polygon with 40 sides?

(Student-produced response)

### ***Question 11***

Skill: Nonlinear Functions | Difficulty: Medium

[PLACEHOLDER: Table with columns: x | h(x). Rows: 0, 12 | 1, 14 | 2, 20. Note: 3^0=1, so h(0)=1+11=12 doesn't work cleanly. Use: 0→12, 1→14, 2→20. Actually use: x: 0→13, 1→15, 2→21 so h(x)=3^x+12 works: 1+12=13, 3+12=15, 9+12=21.]

For the function *h*, the table gives three values of *x* and their corresponding values of $h(x)$. Which equation could define *h*?

A) $h(x) = 3^x + 12$

B) $h(x) = 3^x + 11$

C) $h(x) = 3^x + 3$

D) $h(x) = 3^x$

### ***Question 12***

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Medium

$f(x) = 5(x - 2)(x - 7)(x - 13)$

If the given function *f* is graphed in the xy-plane, where $y = f(x)$, what is the x-coordinate of an x-intercept of the graph?

(Student-produced response)

## *Question 13*

Skill: Systems of Two Linear Equations in Two Variables | Difficulty: Medium

$x + 5y = 23$
$3x + y = 5$

The solution to the given system of equations is ($x$, $y$). What is the value of $4x + 6y$?

A) 14
B) 18
C) 28
D) 42

## *Question 14*

Skill: Lines, Angles, and Triangles | Difficulty: Medium

In triangle **GHJ**, the measure of angle **G** is 52° and the measure of angle **H** is 93°. In triangle **KLM**, the measure of angle **K** is 52° and the measure of angle **L** is 93°. Which of the following additional pieces of information is needed to determine whether triangle **GHJ** is similar to triangle **KLM**?

A) The measure of angle **J**
B) The measure of angle **M**
C) The measure of angle **J** and the measure of angle **M**
D) No additional information is needed.

## *Question 15*

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Hard

$x^2 + y = 25$
$-x + y = 1$

The solutions to the given system of equations are of the form ($x$, $y$). What is a possible value of $x$?

A) -6
B) -4
C) -1
D) 24

## *Question 16*

Skill: Nonlinear Functions | Difficulty: Hard

In the xy-plane, the graph of function $f$, where $y = f(x)$, has exactly 6 x-intercepts. One of these x-intercepts is (13, 0). The rational function $g$ is defined by $g(x) = f(x) / (x – 13)$. In the xy-plane, how many x-intercepts does the graph of $y = g(x)$ have?

(Student-produced response)

### *Question 17*

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Medium

$(x – 5)(x – 11) / (x – 3) = 0$

What is the sum of the solutions to the given equation?

A) 6
B) 11
C) 16
D) 22

### *Question 18*

Skill: Linear Equations in Two Variables | Difficulty: Hard

In the xy-plane, line *t* passes through the point (0, 0) and is parallel to the line represented by the equation y = 38x + 7. If line *t* also passes through the point (5, *d*), what is the value of *d*?

(Student-produced response)

### *Question 19*

Skill: Area and Volume / Circles | Difficulty: Medium

A circle has center *P*, and points *C* and *D* lie on the circle. The measure of arc **CD** is 90°, and the length of this arc is 5 inches. What is the circumference, in inches, of the circle?

A) 5
B) 10
C) 15
D) 20

### *Question 20*

Skill: Ratios, Rates, Proportional Relationships | Difficulty: Medium

While the mass of an object is the same everywhere, the weight of an object is not the same on different planets. An object has a weight of 150.00 pounds on Earth and a weight of 170.25 pounds on Neptune. The object's weight on Jupiter is 236.4% of its weight on Earth. If the object's weight on Neptune is *x*% of its weight on Jupiter, which of the following is closest to the value of *x*?

A) 42.34
B) 71.83
C) 270.09
D) 480.10

### *Question 21*

Skill: Systems of Two Linear Equations in Two Variables | Difficulty: Hard

$-x - wy = -245$
$3x - wy = 73$

In the given system of equations, $w$ is a constant. In the xy-plane, the graphs of these equations intersect at the point $(q, 14)$, where $q$ is a constant. What is the value of $w$?

(Student-produced response)

### *Question 22*

Skill: Right Triangles and Trigonometry | Difficulty: Hard

[PLACEHOLDER: Right triangle with vertices A (top), B (bottom-left), C (bottom-right). Right angle at C. Hypotenuse AB = 130. Angle at B = 25°. Figure not drawn to scale. The side opposite to angle B is AC and the side adjacent to angle B is BC.]

In the right triangle shown, what is the value of tan $B$?

(Student-produced response)

# SECTION 2: MATH

## Module 2 — Easier Path (22 Questions)

(~70% Easy/Medium, ~30% Hard)

### *Question 1*

Skill: Linear Functions | Difficulty: Easy

The function *p* is defined by $p(x) = 24 - x$. What is the value of $p(6)$?

A) 4
B) 18
C) 24
D) 30

### *Question 2*

Skill: Nonlinear Functions | Difficulty: Easy

[PLACEHOLDER: Graph showing a downward-opening parabola. X-axis labeled 'Price (dollars)' from 0 to 80. Y-axis labeled 'Profit (dollars)' from 0 to 1,800. The parabola has a vertex at approximately (30, 1,600), crosses x-axis near x=5 and x=55.]

The graph gives the estimated profit, in dollars, for a bakery when the price of a loaf of bread is *x* dollars. What does the y-value of the vertex of the graph represent in this context?

A) The price, in dollars, of a loaf of bread
B) The minimum estimated profit, in dollars
C) The maximum estimated profit, in dollars
D) The estimated profit, in dollars, when the price of a loaf of bread is 0 dollars

### *Question 3*

Skill: Linear Equations in Two Variables | Difficulty: Easy

[PLACEHOLDER: Table with columns: x | y. Rows: 0, 70 | 1, 78 | 2, 86.]

For a linear relationship between *x* and *y*, the table gives three values of *x* and their corresponding values of *y*. Which equation represents this relationship?

A) $y = 8x + 70$
B) $y = x + 78$
C) $y = x + 8$
D) $y = 8x$

### *Question 4*

Skill: Area and Volume | Difficulty: Easy

A circle has a radius of 17 meters. What is the area, in square meters, of the circle?

A) $17\pi/2$
B) $17\pi$
C) $34\pi$
D) $289\pi$

### *Question 5*

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Medium

$p = \sqrt{(10c - 36)}$

The given equation relates the variables *p* and *c*, where *c* is positive. Which equation correctly expresses *c* in terms of *p*?

A) $c = (p + 36)^2 / 10$
B) $c = p^2/10 + 36/10$
C) $c = p/18 + 36/18$
D) $c = \sqrt{(p + 36)} / 10$

### *Question 6*

Skill: Linear Functions | Difficulty: Easy

The function *f* is defined by $f(x) = -76x$. What is the slope of the graph of $y = f(x)$ in the xy-plane?

A) -76
B) -1/76
C) 1
D) 76

### *Question 7*

Skill: Linear Functions | Difficulty: Medium

A linear model estimates the population of a town from 1995 to 2019. The model estimates the population was 48 thousand in 1995, 205 thousand in 2015, and *x* thousand in 2019. To the nearest whole number, what is the value of *x*?

(Student-produced response)

### *Question 8*

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Medium

$w^2 + 6w - 40 = 0$

Which of the following is a solution to the given equation?

A) $4 - 2\sqrt{14}$
B) $2\sqrt{14}$
C) $\sqrt{14}$
D) $-6 + 2\sqrt{14}$

## *Question 9*

Skill: Lines, Angles, and Triangles | Difficulty: Hard

[PLACEHOLDER: Figure showing triangle FHI with point R on segment FH. Segment RG is drawn parallel to segment HI. Angle at I = 31°. Angle FRG = 68°. Angles p and r are marked at vertices F and H respectively. Figure not drawn to scale.]

In the figure shown, point *R* lies on segment **FH**, and segment **RG** is parallel to segment **HI**. The measure of angle *I* is 31°, and the measure of angle **FRG** is 68°. What is the value of $p - r$?

(Student-produced response)

## *Question 10*

Skill: Nonlinear Functions | Difficulty: Medium

On March 1, there were 185 views of a blog post on a website. Every 3 days after March 1, the number of views of the blog post had increased by 50% of the number of views 3 days earlier. The function *f* gives the predicted number of views *x* days after March 1. Which equation defines *f*?

A) $f(x) = 185(0.50)^{x/3}$
B) $f(x) = 185(0.50)^{3x}$
C) $f(x) = 185(1.50)^{x/3}$
D) $f(x) = 185(1.50)^{3x}$

## *Question 11*

Skill: Linear Inequalities in One or Two Variables | Difficulty: Medium

[PLACEHOLDER: Coordinate plane showing a shaded region. The boundary line passes through (0, 0) and has a negative slope. The shaded region is below and to the right of the line. The line appears to pass through points like (8, -8) and (-8, 8), suggesting the boundary is y = -x, so sy < 96 with the line being y = 96/s * (1/x). Actually: boundary line passes through (-8, 2) and (8, -2) approximately. The shaded region is below the line. Grid from -8 to 8 on x-axis, -10 to 2 on y-axis.]

The shaded region shown represents the solutions to the inequality $sy < 96$, where s is a constant. What is the value of s?

A) 12
B) 6
C) -6
D) -12

## *Question 12*

Skill: Linear Equations in One Variable | Difficulty: Medium

How many solutions does the equation $6(12x - 18) = -18(6 - 4x)$ have?

A) Exactly one
B) Exactly two
C) Infinitely many
D) Zero

### *Question 13*

Skill: Nonlinear Functions | Difficulty: Medium

$f(x) = 5{,}200(0.45)^{x/12}$

The function $f$ gives the value, in dollars, of a certain piece of machinery after $x$ months of use. If the value of the machinery decreases each year by $p$% of its value the preceding year, what is the value of $p$?

A) 4
B) 12
C) 45
D) 55

### *Question 14*

Skill: Linear Functions | Difficulty: Hard

A gardener's initial count of aphids on a certain rosebush was 800. The gardener set a goal to reduce the population to 200 aphids. The gardener uses a model that predicts the population begins at 800 and decreases by 90 aphids per week in the first three weeks, and then decreases by 150 aphids per week thereafter. According to this model, at the end of week $w$ after the initial count, where $w > 3$, which of the following functions gives the predicted number of aphids still needed to reach the gardener's goal?

A) $p(w) = 200 - 150w$
B) $p(w) = 330 + 150w$
C) $p(w) = 680 - 150w$
D) $p(w) = -80 + 150w$

### *Question 15*

Skill: Linear Equations in Two Variables | Difficulty: Medium

[PLACEHOLDER: Coordinate plane showing a line with a positive slope. The line crosses the y-axis at (0, 4) and the x-axis at (-3, 0). Grid visible from -1 to 9 on x-axis and -9 to 4 on y-axis.]

What is an equation of the graph shown?

A) $4x - 3y = 12$
B) $4x - 3y = -12$
C) $3x - 4y = 12$
D) $3x - 4y = -12$

### *Question 16*

Skill: Equivalent Expressions | Difficulty: Hard

Which expression is a factor of $25x^2(x + 8) - 25(x + 8)^3$?

A) $-10x + 80$
B) $5x + 8$
C) $8x + 8$
D) $20x - 80$

### *Question 17*

Skill: Systems of Two Linear Equations in Two Variables | Difficulty: Hard

$24x = 1{,}500y - 2{,}400$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

A) $(1/24)x = 2y$

B) $(1/24)x = 50y - 100$

C) $x = 2y$

D) $x = 50y - 100$

### *Question 18*

Skill: Nonlinear Functions | Difficulty: Hard

The exponential function *f* is defined by $f(x) = a \cdot b^x$, where *a* and *b* are positive constants. If $f(s) = t$ and $f(s + 1) = t \cdot 0.73t / t$, where *s* and *t* are constants, what is the value of *b*?

A) 0.27

B) 0.73

C) 1.27

D) 1.73

### *Question 19*

Skill: Ratios, Rates, Proportional Relationships | Difficulty: Hard

A tank has a volume of 0.72 cubic meters. Which of the following is closest to this volume, in gallons?
(Note: 100.00 centimeters = 1 meter. Use 3,785.41 cubic centimeters = 1.00 gallon.)

A) 0.07

B) 5.27

C) 190.19

D) 756.14

### *Question 20*

Skill: Nonlinear Functions | Difficulty: Hard

Function *g* is a quadratic function. The graph of $y = g(x)$ in the xy-plane has a vertex at (5, –8), contains the point (4, –5), and has a y-intercept at (0, *a*). The graph of $y = 6 \cdot g(x)$ has a y-intercept at (0, *b*). What is the positive difference between *a* and *b*?

(Student-produced response)

### *Question 21*

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Hard

$-x^2 + cx - 81 = 0$

In the given equation, *c* is an integer. The equation has no real solutions. What is the least possible value of *c*?

(Student-produced response)

### *Question 22*

Skill: Ratios, Rates, Proportional Relationships | Difficulty: Hard

A biologist investigated two species of beetles: a predator and its prey. At the start of a month, there was an equal number of the two species. At the end of the month, the number of prey had increased by 3,200% of the number of prey at the start of the month, and the number of predators had increased by 220% of the number of predators at the start of the month. The number of prey at the end of the month was $p$% greater than the number of predators at the end of the month. What is the value of $p$?

(Student-produced response)

# SECTION 2: MATH

## Module 2 — Harder Path (22 Questions)

(~30% Easy/Medium, ~70% Hard)

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### *Question 1*

Skill: Linear Functions | Difficulty: Easy

The function *h* is defined by $h(x) = 32 - x$. What is the value of $h(5)$?

A) 5
B) 16
C) 27
D) 37

### *Question 2*

Skill: Nonlinear Functions | Difficulty: Medium

[PLACEHOLDER: Graph showing an upward-opening parabola. X-axis labeled 'Production quantity (units)' from 0 to 100. Y-axis labeled 'Cost (dollars)' from 0 to 2,500. The parabola has a vertex at approximately (40, 400), meaning the minimum cost occurs at 40 units.]

The graph gives the estimated cost, in dollars, for a manufacturer when the production quantity is *x* units. What does the y-value of the vertex of the graph represent in this context?

A) The production quantity, in units
B) The minimum estimated cost, in dollars
C) The maximum estimated cost, in dollars
D) The estimated cost, in dollars, when the production quantity is 0 units

### *Question 3*

Skill: Linear Equations in Two Variables | Difficulty: Easy

[PLACEHOLDER: Table with columns: x | y. Rows: 0, 45 | 1, 57 | 2, 69.]

For a linear relationship between *x* and *y*, the table gives three values of *x* and their corresponding values of *y*. Which equation represents this relationship?

A) $y = 12x + 45$
B) $y = x + 57$
C) $y = x + 12$
D) $y = 12x$

### *Question 4*

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Hard

$p = \sqrt{(14c - 50)}$

The given equation relates the variables *p* and *c*, where *c* is positive. Which equation correctly expresses *c* in terms of *p*?

- A) $c = (p + 50)^2 / 14$
- B) $c = p^2/14 + 50/14$
- C) $c = p/22 + 50/22$
- D) $c = \sqrt{(p + 50)} / 14$

### *Question 5*

Skill: Linear Functions | Difficulty: Medium

The function *f* is defined by $f(x) = -54x$. What is the slope of the graph of $y = f(x)$ in the xy-plane?

- A) -54
- B) -1/54
- C) 1
- D) 54

### *Question 6*

Skill: Linear Functions | Difficulty: Hard

A linear model estimates the enrollment at a university from 1998 to 2022. The model estimates the enrollment was 63 thousand in 1998, 231 thousand in 2018, and *x* thousand in 2022. To the nearest whole number, what is the value of *x*?

(Student-produced response)

### *Question 7*

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Hard

$w^2 + 10w - 39 = 0$

Which of the following is a solution to the given equation?

- A) $5 - 2\sqrt{16}$
- B) $2\sqrt{16}$
- C) $\sqrt{16}$
- D) $-5 + 8$

### ***Question 8***

Skill: Lines, Angles, and Triangles | Difficulty: Hard

[PLACEHOLDER: Figure showing triangle EGH with point M on segment EG. Segment MN is drawn parallel to segment GH. Angle at H = 29°. Angle EMN = 72°. Angles a and b are marked at vertices E and G respectively. Figure not drawn to scale.]

In the figure shown, point *M* lies on segment **EG**, and segment **MN** is parallel to segment **GH**. The measure of angle *H* is 29°, and the measure of angle **EMN** is 72°. What is the value of $a - b$?

(Student-produced response)

### ***Question 9***

Skill: Nonlinear Functions | Difficulty: Hard

On April 1, there were 310 subscribers to a newsletter. Every 4 days after April 1, the number of subscribers had increased by 40% of the number of subscribers 4 days earlier. The function *f* gives the predicted number of subscribers *x* days after April 1. Which equation defines *f*?

A) $f(x) = 310(0.40)^{x/4}$
B) $f(x) = 310(0.40)^{4x}$
C) $f(x) = 310(1.40)^{x/4}$
D) $f(x) = 310(1.40)^{4x}$

### ***Question 10***

Skill: Linear Inequalities in One or Two Variables | Difficulty: Hard

[PLACEHOLDER: Coordinate plane showing a shaded region. The boundary line has a negative slope passing through the origin area. The shaded region is below the boundary line. The line appears to pass through (-8, 2) and (8, -2) approximately, with the shaded region below it. Grid from -8 to 8 on x-axis, -10 to 2 on y-axis.]

The shaded region shown represents the solutions to the inequality $ty < 144$, where t is a constant. What is the value of t?

A) 18
B) 9
C) -9
D) -18

### ***Question 11***

Skill: Linear Equations in One Variable | Difficulty: Hard

How many solutions does the equation $5(15x – 25) = -25(5 – 3x)$ have?

A) Exactly one
B) Exactly two
C) Infinitely many
D) Zero

### *Question 12*

Skill: Nonlinear Functions | Difficulty: Hard

$f(x) = 8{,}400(0.28)^{x/12}$

The function *f* gives the value, in dollars, of a certain instrument after *x* months of use. If the value of the instrument decreases each year by *p*% of its value the preceding year, what is the value of *p*?

A) 7
B) 12
C) 28
D) 72

### *Question 13*

Skill: Linear Functions | Difficulty: Hard

A beekeeper's initial observation of the population of a certain hive was 900 bees. The beekeeper set a goal to increase the population to 2,400 bees. The beekeeper uses a model that predicts the population begins at 900 and increases by 100 bees per week in the first three weeks, and then increases by 160 bees per week thereafter. According to this model, at the end of week *w* after the initial observation, where $w > 3$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

A) $p(w) = 2{,}400 – 160w$
B) $p(w) = 2{,}220 + 160w$
C) $p(w) = 1{,}320 – 160w$
D) $p(w) = -180 + 160w$

### *Question 14*

Skill: Linear Equations in Two Variables | Difficulty: Hard

[PLACEHOLDER: Coordinate plane showing a line with a positive slope. The line crosses the x-axis at (4, 0) and the y-axis at (0, -7). Grid visible.]

What is an equation of the graph shown?

A) $7x – 4y = 28$
B) $7x – 4y = -28$
C) $4x – 7y = 28$
D) $4x – 7y = -28$

### *Question 15*

Skill: Equivalent Expressions | Difficulty: Hard

Which expression is a factor of $49x^2(x + 6) – 49(x + 6)^3$?

A) $-14x + 84$
B) $7x + 6$
C) $6x + 6$
D) $28x – 84$

### *Question 16*

Skill: Systems of Two Linear Equations in Two Variables | Difficulty: Hard

$36x = 2{,}100y – 4{,}200$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

A) $(1/36)x = 2y$

B) $(1/36)x = 175/3 \cdot y – 350/3$

C) $x = 2y$

D) $x = 175/3 \cdot y – 350/3$

### *Question 17*

Skill: Nonlinear Functions | Difficulty: Hard

The exponential function *f* is defined by $f(x) = a \cdot b^x$, where *a* and *b* are positive constants. If $f(s) = t$ and $f(s + 1) = 0.91t$, where *s* and *t* are constants, what is the value of *b*?

A) 0.09

B) 0.91

C) 1.09

D) 1.91

### *Question 18*

Skill: Ratios, Rates, Proportional Relationships | Difficulty: Hard

A reservoir has a volume of 0.95 cubic meters. Which of the following is closest to this volume, in gallons? (Note: 100.00 centimeters = 1 meter. Use 3,785.41 cubic centimeters = 1.00 gallon.)

A) 0.10

B) 3.60

C) 250.96

D) 1,006.35

### *Question 19*

Skill: Nonlinear Functions | Difficulty: Hard

Function *g* is a quadratic function. The graph of $y = g(x)$ in the xy-plane has a vertex at $(3, –10)$, contains the point $(2, –7)$, and has a y-intercept at $(0, a)$. The graph of $y = 4 \cdot g(x)$ has a y-intercept at $(0, b)$. What is the positive difference between *a* and *b*?

(Student-produced response)

### *Question 20*

Skill: Nonlinear Equations in One Variable and Systems | Difficulty: Hard

$-x^2 + dx – 49 = 0$

In the given equation, *d* is an integer. The equation has no real solutions. What is the least possible value of *d*?

(Student-produced response)

## *Question 21*

Skill: Ratios, Rates, Proportional Relationships | Difficulty: Hard

A researcher investigated two species of fish: a predator and its prey. At the start of a season, there was an equal number of the two species. At the end of the season, the number of prey had increased by 1,800% of the number of prey at the start of the season, and the number of predators had increased by 150% of the number of predators at the start of the season. The number of prey at the end of the season was $p$% greater than the number of predators at the end of the season. What is the value of $p$?

(Student-produced response)

## *Question 22*

Skill: Systems of Two Linear Equations in Two Variables | Difficulty: Hard

$-x - wy = -312$
$4x - wy = 88$

In the given system of equations, $w$ is a constant. In the xy-plane, the graphs of these equations intersect at the point ($q$, 11), where $q$ is a constant. What is the value of $w$?

(Student-produced response)

# QUESTION COUNT VERIFICATION

- S1 M1: 27 | S1 M2 Easier: 27 | S1 M2 Harder: 27
- S2 M1: 22 | S2 M2 Easier: 22 | S2 M2 Harder: 22
- TOTAL: 147



Easier Module 2: ~70% Easy/Medium, ~30% Hard
Harder Module 2: ~30% Easy/Medium, ~70% Hard